# Delineated Analysis of Robotic Process Automation Tools

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***Abstract*- In this age and time when celerity is expected out of all the sectors of the country, the speed of execution of various processes and hence efficiency, becomes a prominent factor. To facilitate the speeding demands of these diverse platforms, Robotic Process Automation (RPA) is used. Robotic Process Automation can expedite back-office tasks in commercial industries, remote management tasks in IT industries and conservation of resources in multiple sectors. To implement RPA, many software platforms are developed and the objective of this paper will be providing an analysis of three of the leading RPA platforms namely, UiPath Studio, Automation Anywhere and Blue Prism. Our analysis will help the commercial industries- not only the front-office but also for back-office personnel; to determine which platform is the most eligible to use.**

***Keywords: Robotic Process Automation, UiPath, Blue Prism, Automation Anywhere, Structured Analysis***

## I. INTRODUCTION

To automatically control manual or logical processes, a technique known as Process Automation is used. The current developments in technology have made it possible to introduce industry automation systems into all possible fields. Process Automation is broadly categorized into two parts – Hard Automation and Soft Automation.[1]

Hard Automation (fixed automation) refers to a machine or robot designed to perform a specific yet repetitive task whereas Soft Automation (flexible automation) is a developed version of the former since it allows programming of different tasks as per the need of the products.

Robotic Process Automation (RPA) comes under Soft Automation.[2]

Robotic Process Automation is a software that can be programmed to automate a variety of manual processes that human workers do, to reduce the burden of recurring tasks.

The software bot is taught a workflow with multiple steps and applications such as reading data, writing data, numeric calculations, modification and renewal of data, and so on. Currently, there are multiple tools being used for RPA such as UiPath Studio, Automation Anywhere, Blue Prism, Workfusion, Redwood, Kryon, etc. [3]

The outline of this paper can be described as: Part II explains the history and description of the top 3 tools in the RPA market. Part III provides a comprehensive and detailed comparison of the three respective platforms. Part IV shows the implementation of a robot that we created through UiPath Studio. Part V inculcates the future scope and Part VI has the conclusion of our analysis.

## II. SYSTEM OVERVIEW

### *A. UiPath*

UiPath dates back to 2005, in what was back then an outsourcing company. With a strong demand from the market, they saw the need for RPA (Robotic Process Automation), and hence decided to start building an industry standard platform for training and orchestrating software robots. [2]

Their source code is being is used on millions of machines around the world, embedded in different products and companies with activities ranging from document management to call centre, healthcare, financial, data extraction and migration, process automation, or API enablement.

### *B. Automation Anywhere*

In 2010, the company, Tethys Solutions, LLC, re-branded itself as Automation Anywhere, Inc. The company's products are designed to allow execution of automated business and IT processes across multiple machines, allowing for variations in systems, application loading times and Internet speeds. It is available in a Server edition that allows users to develop automation processes with centralized security, user management, collaboration, and deployment and backup. [4]

### *C. Blue Prism*

Blue Prism was founded in 2001 by a group of process automation experts to develop a technology that could be used to improve the efficiency and effectiveness of organizations. Their focus was on the white-collar back office where they recognized an enormous unfulfilled need for automation. [5]

## III. COMPARATIVE STUDY

Out of the plethora of grounds to compare the top three RPA platforms, in Table 1, we have enunciated the most prominent factors. The basic parameter suggests whether front office and back office automation is possible or not. The requirement to automate the very initial stages of an industry forms the initial layer of success for any automation tool or

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platform. [6] Further ahead, the Script Based Designer and Visual Process Designers come into context where, we check whether the particular tool provides graphic user interface or not. The openness of a platform states whether the material to understand the tool, learn how to operate it and practice various applications on it; is provided for all or not. Macro Recorders help in faster coding and design implementation. The Control through Coding criteria is important since it suggests how efficient it would be for a user to control the function of the tool and the bots the tool implements. Whether the execution of automated test cases on remote machines is possible or not, curates the possibility of the tool being secure. If at all, a tool can fulfil the above criteria without compromising security, a massive improvement in this field would be made. [7] For the last parameter, the extent of a tool's future scope determines how useful it would be in the long run when other technologies would be far ahead. UiPath dominates in this category since it's forever adaptive coded algorithms make an infinite future scope possible, unlike the other two. [8]

TABLE I. Feature Based Comparative Study

| **Parameters** | **UiPath** | **Blue Prism** | **Automation Anywhere** |
|---|---|---|---|
| Front Office/ Attended Automation | Yes | No | Yes |
| Back Office/ Unattended Automation | Yes | Yes | Yes |
| Script Based Designer | No | No | Yes |
| Visual Process Designer | Yes | Yes | Yes but, is more script based. |
| Openness of the Platform | Yes, has free forums and tutorials. | Yes but, all the forums are commercial. | Yes but, all the forums are commercial.[9] |
| Macro Recorder for Process Mapping | Yes | No, due to their rather Outdated technology. | Yes |
| Control through Coding | No | Yes | Yes |
| Execution of Automated Test Cases on Remote Machines | No | No | Yes |
| Future Scope | Indefinite | Comparatively less | Comparatively less |

Provision of a simple factual analysis might not be enough to sufficiently convince the backbones of various industries. This necessitates a comparison based on the technical aspects of the tools. In Table 2, we have cumulated the performance evaluation of the different tools out of 5, based on the different technological categories. This data has been amassed based on industrial reviews of RPA tools and our own implementations on UiPath since that was the only tool easily available at the current stage of our analysis. [10] As Automation Anywhere has a script based design too, the bot development and core functioning can be implemented with higher accuracy However, the number of users might decrease precisely because it is script based. Control room, System Management, Reporting and Resilience incorporate the operating requirements and factors of the tools and UiPath and Blue Prism perform well in these areas. The RPA Analytics rating defines the analytical potential of that RPA tool. [11] The architecture of Automations Anywhere is the most commendable since it is very detailed. Blue Prism provides the most security however UiPath and Automation Anywhere do fairly well in this criterion. However, overall UiPath triumphs all. [12]

TABLE II. Comparative Study on Technical Aspects

| **Technology Category** | **UiPath** | **Blue Prism** | **Automation Anywhere** |
|---|---|---|---|
| Bot Development and Core Functions | 3.28 | 2.56 | 3.74 |
| Control Room, System Management, Reporting and Resilience | 3.84 | 3.84 | 2.84 |
| RPA Analytics | 3.68 | 2.00 | 3.68 |
| Architecture | 4.00 | 3.68 | 4.34 |
| Deployment, Governance and Security | 3.68 | 4.00 | 3.68 |
| Total RPA Technology Score | 3.67 | 3.20 | 3.64 [13] |

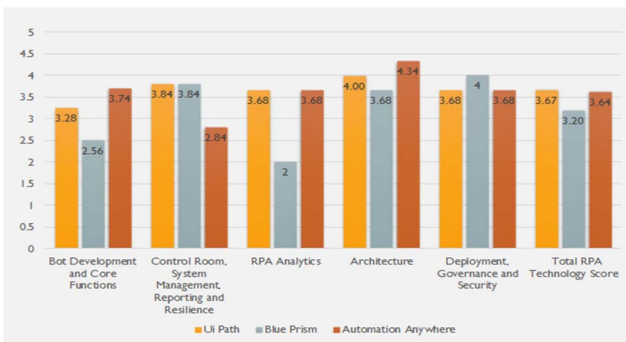


Fig. 3.1: Graphical Representation of Technical Aspects

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## IV. IMPLEMENTATION

We implemented a small bot for the evaluation of the tool that obtains the best results after a careful technical analysis; UiPath. This UiPath bot is a small-scale bot which facilitates the idea of making data recording in educational institutions easy. The UiPath bot operates on the data of four different files to create a new file which has the accumulated data from all the files. The first file contains the records of the students including their personal information. The rest of the three files contain information regarding the marks and roll number and marks of three subjects namely, Maths, Science and English. The data from all these files is accumulated into a fourth file which provides the complete information about the concerned students. The bot also provides an option of calculating the percentage which can later be replaced with various other mathematical, statistical and logical operations.

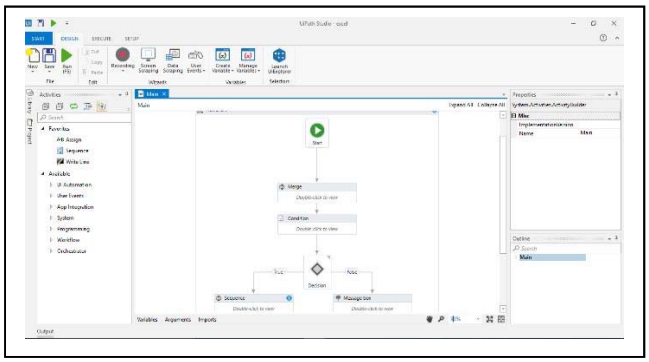
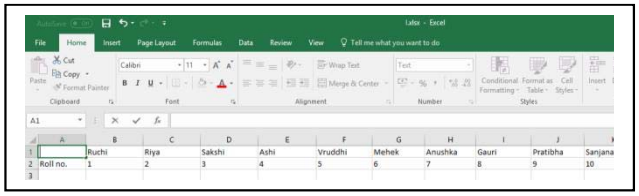


Fig. 4.1: An overview of UiPath Studio

The functioning of the bot is as follows:

### A. Step 1

In this test case, an excel file called "List.xlsx" is made that contains the names and roll numbers of 10 students.

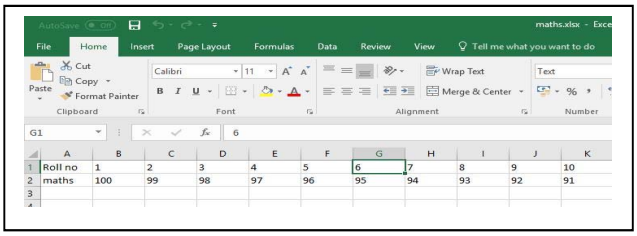


| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | Ruchi | Riya | Sakshi | Ashi | Vruddhi | Mehek | Anushka | Gauri | Pratibha |
| 2 | Roll no. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
| 3 | | | | | | | | | | |

Fig. 4.2: List.xlsx Excel file

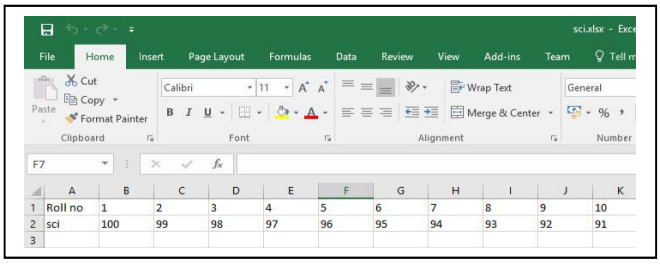
### B. Step 2

Three different files called "Maths.xlsx", "Science.xlsx" and "English.xlsx" are made that have records of the marks of all the ten students in the respective subjects.



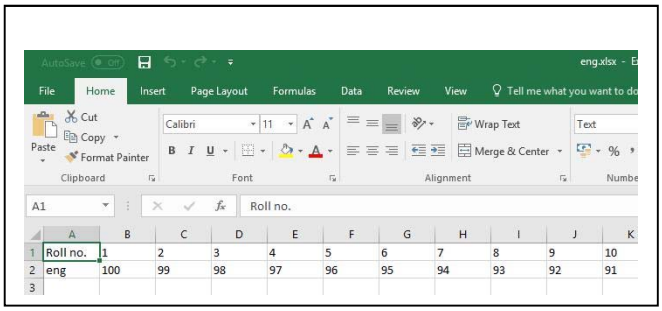
| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Roll no | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | maths | 100 | 99 | 98 | 97 | 96 | 95 | 94 | 93 | 92 | 91 |
| 3 | | | | | | | | | | | |

Fig. 4.3: Maths.xlsx Excel file



| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Roll no | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | sci | 100 | 99 | 98 | 97 | 96 | 95 | 94 | 93 | 92 | 91 |
| 3 | | | | | | | | | | | |

Fig. 4.4: Science.xlsx Excel file



| | A | B | C | D | E | F | G | H | I | J | K |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | Roll no. | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
| 2 | eng | 100 | 99 | 98 | 97 | 96 | 95 | 94 | 93 | 92 | 91 |
| 3 | | | | | | | | | | | |

Fig 4.5: English.xlsx Excel file

### C. Step 3

A bot is created that accumulates the data from these four files. The 'Merge' section is responsible for merging the data together as specified. The 'Condition' section is responsible for asking the user whether they want to calculate the percentage of the students or not. In the end, the 'Decision' section acts as an 'If Else' statement and directs the further functioning of the program based on the answer that the user has provided.

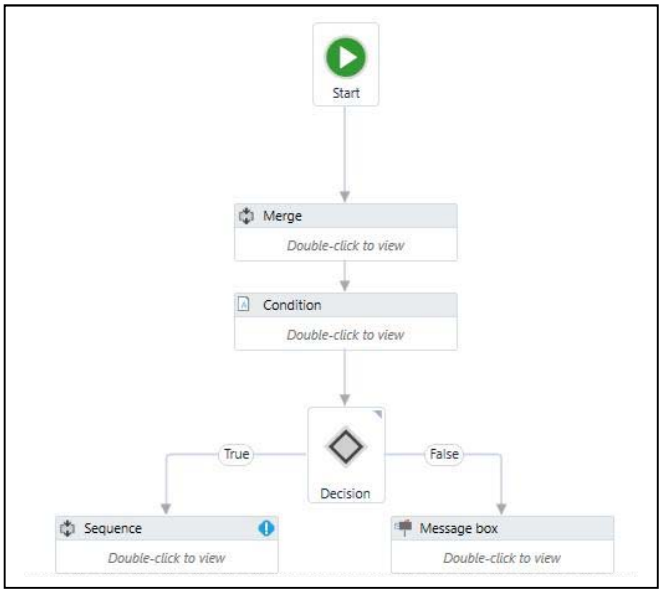



Fig. 4.6: An overview of the bot

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D. *Step 4*

The working of the blocks is as follows:

1. *Merge*

It is responsible for merging the four different files together and accumulating their data.

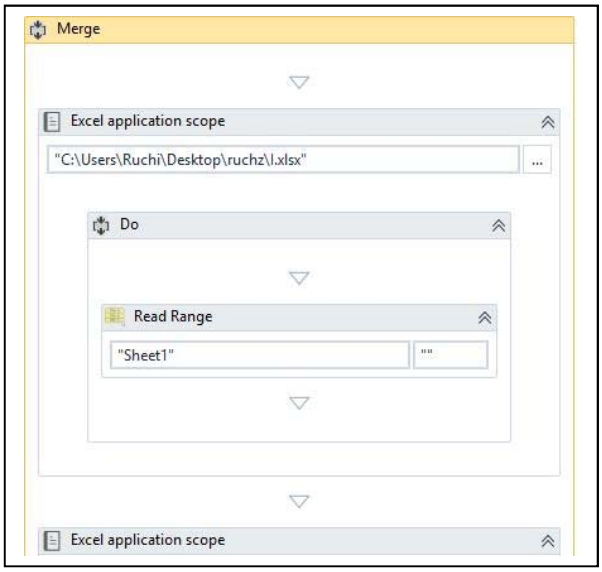


Fig. 4.7: The Merge Block

2. *Conditions*

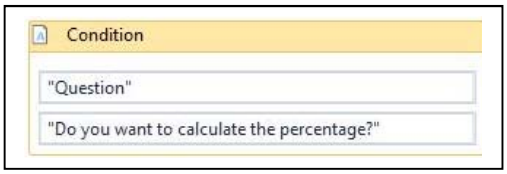


Fig. 4.8: The Condition Block

It is responsible for asking the user if he/she wishes to calculate the percentage of all the students.

3. *Decision*

(i) *True*

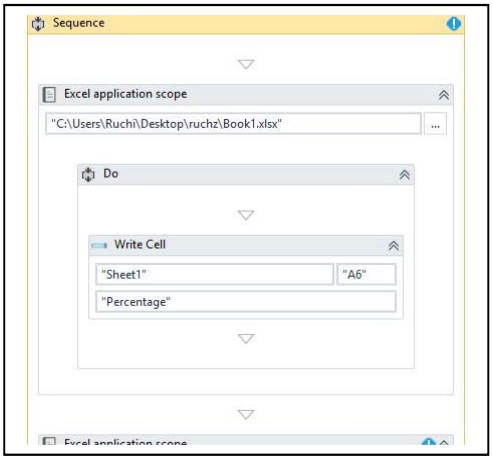


Fig. 4.9: The Decision Block (True)

It is responsible for calculating the cumulative percentage from all three subjects and then displaying the result in the final excel sheet; for every student.

(i) *False*

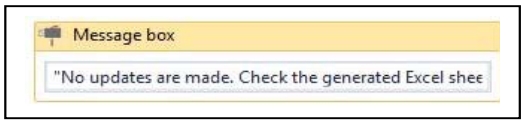


Fig. 4.10: The Decision Block (False)

Since the user chose not to calculate the percentage of the students, the bot directly displays a message box saying that *'No changes were made. Check updated excel sheet.'* And hence, the updated excel sheet is checked.

*E. Step 5*

The excel sheet called "Book1.xlsx" is checked since that is the excel sheet in which the accumulated data is saved in (with/without the percentage).

*F. Result*

"timeStamp": "17:28:40",

"fileName": "Main",

"totalExecutionTimeInSeconds": 28,

"totalExecutionTime": "00:00:28"

The above output from UiPath suggests that the bot was implemented in a total of 28 seconds including 'user input' time.

## V. FUTURE SCOPE

In the future, the scope for UiPath exceeds imaginable boundaries. Credits to this are due to the adaptive algorithms it uses. We will be extending our implementation to develop a full-fledged educational system using UiPath. The system will consist of different modules for various types of users namely- students, teachers, staff and the management committee.

Our system would be implemented using MS SQL which will help store the entire data of the institute in a database. The first module of this system would help students enter their general information and the information about their respective results, and store it in the database so it can be accessed as and when someone authorised wishes to view it. The second module would help teachers to review the results of the students and make grading automated and easier. This module would also include a provision to do multiple mathematical functions if required. Another provision with this module is the system of attendance. The teachers can simply save the data of the attendance in the database and the percentage of attendance would be calculated automatically. The third module would be for the management committee. This would help calculating the salary for the staff based on the number of days they would attend the institution.

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This system is comprised of small tasks that people do every day which take up a lot of time and we don't realise it. These tasks are automated to save up that time which would otherwise be wasted hence, achieving one of the main goals of automation.

## VI. CONCLUSION

With the fast-paced developments, various industries are looking for ways to accommodate multiple tasks and operations in minimum time using less effort. This necessitates the possibility of automation and hence knowing which tool to use for the sake of different sectors based on their type of services becomes a major point of priority. Therefore, in this paper, an overarching study of all the tools has been provided, including their shortcomings and place of optimum use. This paper also specifies that UiPath should be considered the best automation tool out of all of those that are currently in the market and provides a test implementation based on the analytic preview and endless possibilities of processes that can be tentatively automated.

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